

# Observed Variable Models Estimation

## Estimation of Covariance Matrix

If the proposed SEM is correct and the parameters are known then...

$$\Sigma = \Sigma(\theta)$$

In practice we don't know the parameters (but we'll still assume our SEM is correct)

$\mathbf{S}$  is the sample covariance matrix

$$\hat{\Sigma} = \Sigma(\hat{\theta})$$

Our goal is to select values of  $\hat{\theta}$  so that  $\hat{\Sigma}$  is as **close** to  $\mathbf{S}$  as possible

## Maximum Likelihood Goal

Select the **parameter values** which maximize the likelihood that we would observe our sample data

We have observed our data already!

So, we focus on selecting parameter estimates which maximize the *likelihood of observing that data*.

These are the maximum likelihood estimates.

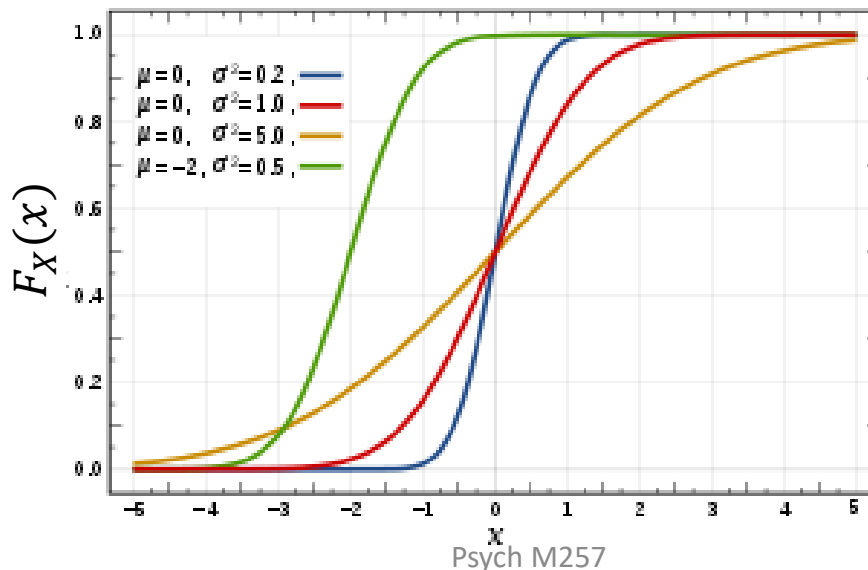
# Cumulative Distribution Functions

All random variables have an associated **cumulative distribution function** (CDF), we'll denote this  $F_X(x)$

A little note on notation. Common a capital letter is meant to represent the “random variable” (i.e.,  $X$ ) and a lower case letter represents a specific value of that random variable (i.e.,  $x$ )

$$F_X(x) = P_X(X \leq x) \text{ for all } x$$

The cumulative distribution function at some value  $x$  is the probability that a random value from  $X$  is less than or equal to  $x$



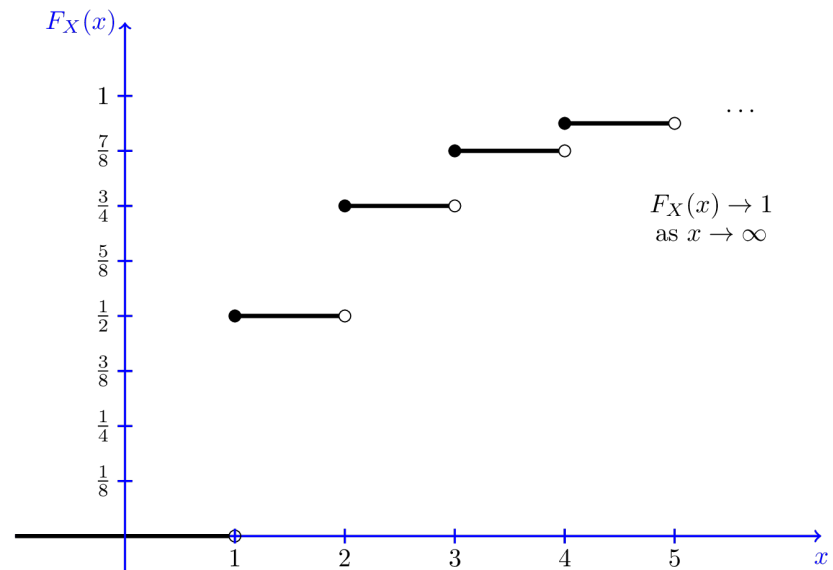
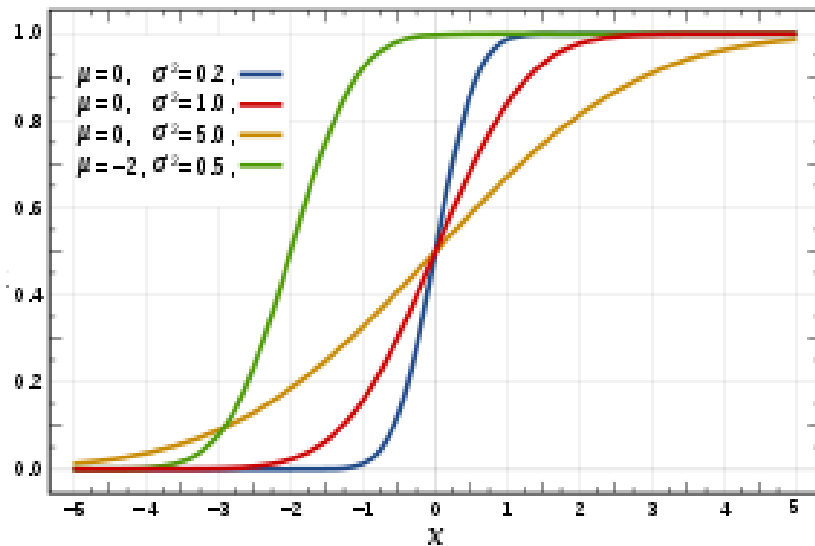
# Cumulative Distribution Functions

## Properties of CDFs:

$$\lim_{x \rightarrow \infty} F_X(x) = 1 \text{ and } \lim_{x \rightarrow -\infty} F_X(x) = 0$$

$F_X(x)$  is non-decreasing

$F_X(x)$  is right continuous (i.e.,  $\lim_{x \downarrow x_0} F_X(x) = F_X(x_0)$ )

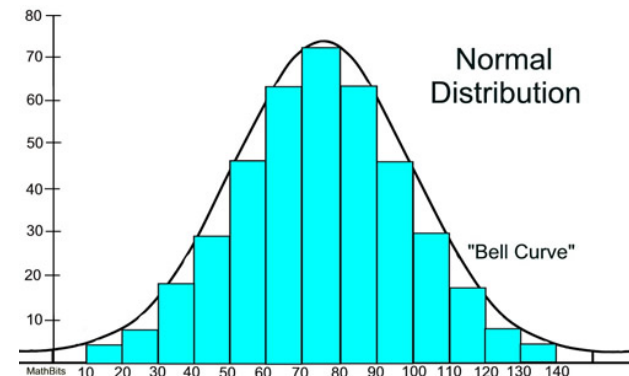
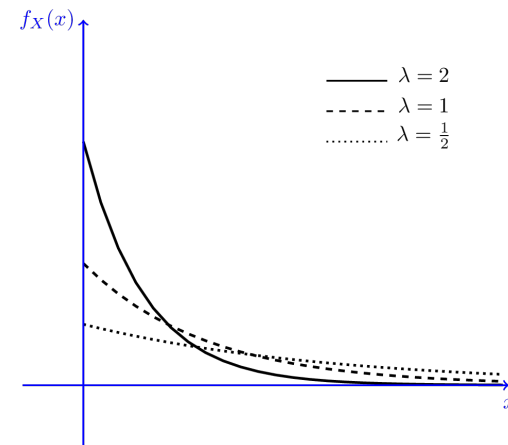
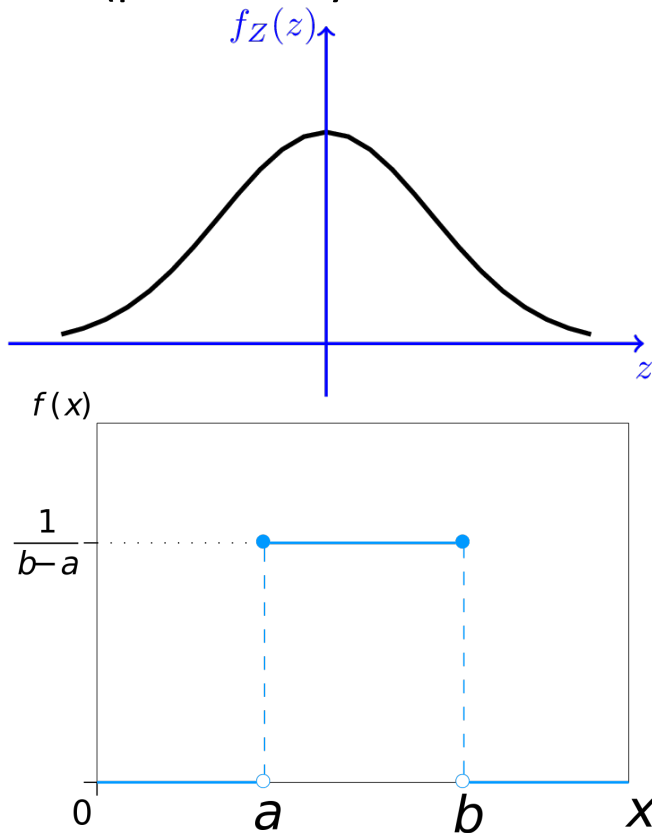


# Probability Density Functions

Probability density functions (pdfs),  $f_X(x)$

pdfs are (probably) the most common way of visualizing a distribution

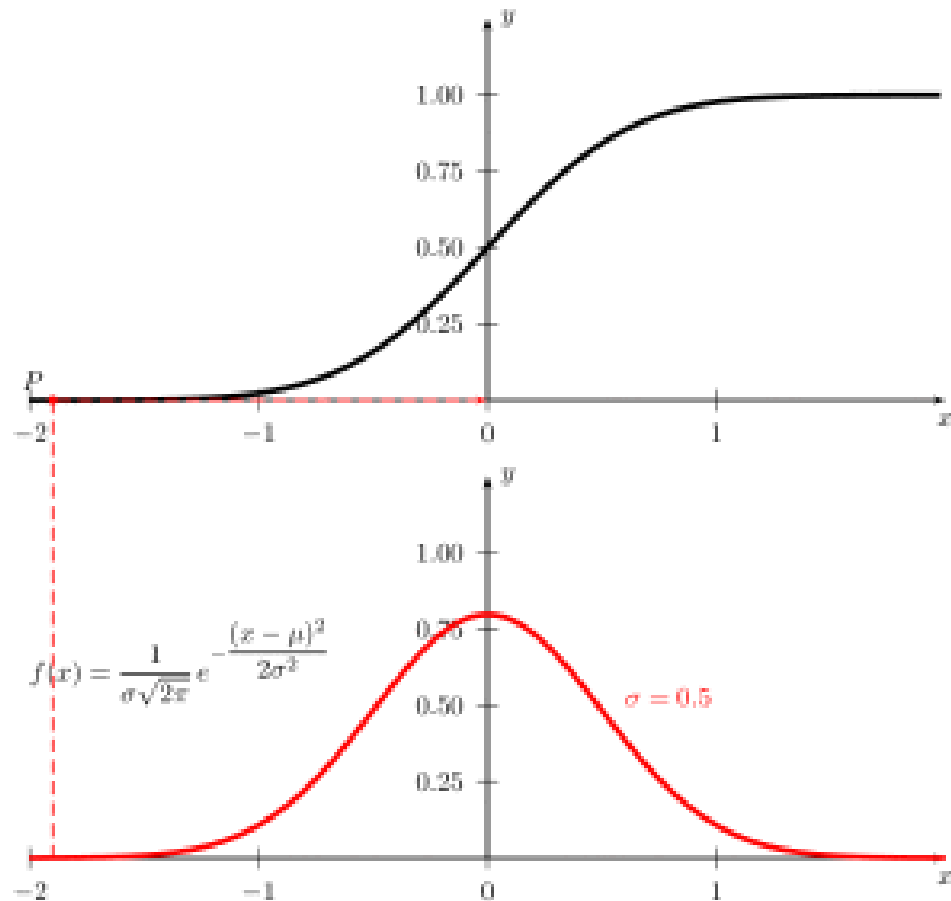
They are the continuous distribution generalization of a histogram (probability mass function)



# Probability Density Functions

$$F_X(x) = \int_{-\infty}^x f_X(t) dt \text{ for all } x$$

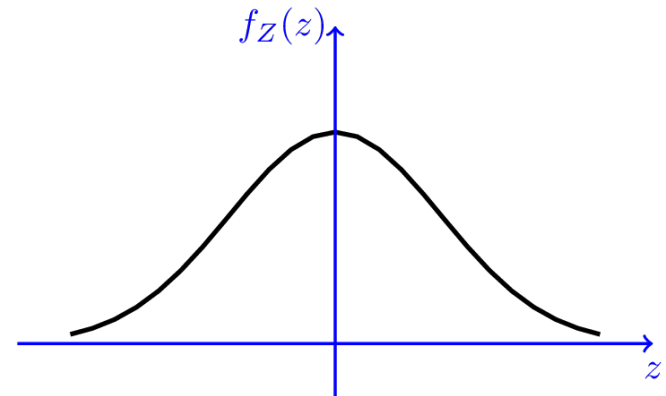
The CDF at  $x$  is the integral of the pdf up to that point on  $x$



## PDFs and Likelihood Functions

Probability density functions (pdfs),  $f_X(x)$

When we treat the parameters as known and the observations as unknown this function is referred to as a probability density function (pdf).



When we treat the parameters as unknown and the observations as known this function is referred to as a **likelihood function**.

The equation does not change, but it specifies how the function is being used.

This is somewhat rooted in the issue in frequentist statistics where we do not discuss the probability of fixed events

Here the data is fixed (and known) and the parameters are fixed (and unknown)

## Coin Flipping

Consider flipping a coin. What if I told you that this coin has one of 3 probabilities, and you need to figure out which it is?

$p = 0.25, 0.50, \text{ or } 0.75$  probability of flipping heads

When we flip a coin, we know that the probability of getting a heads is  $p$

When we flip a coin twice, the probability of getting 2 heads is  $p \times p$

When we flip a coin twice, the probability of getting 2 tails is  $(1 - p) \times (1 - p)$

When we flip a coin twice, the probability of getting 1 heads and 1 tails is  $2(1 - p)(p)$

## Coin Flipping

$$f(x|p) = \binom{2}{x} p^x (1-p)^{(2-x)}$$

$x$  is the number of heads

$p$  is the probability of heads

$\binom{2}{x}$  is read “2 choose  $x$ ” and it is how many ways you can select  $x$  things from 2 things (2 choose 0 is 1, 2 choose 1 is 2, 2 choose 2 is 1)

Our goal is to select a  $p$  which results in the highest  $f(x|p)$  given our  $x$

Consider  $x = 0$

$$f(0|.25) = \binom{2}{0} \cdot .25^0 (1 - .25)^{(2-0)} = 1(1)(.75)^2 = .5625$$

$$f(0|.5) = \binom{2}{0} \cdot .5^0 (1 - .5)^{(2-0)} = 1(1)(.5)^2 = .25$$

$$f(0|.75) = \binom{2}{0} \cdot .75^0 (1 - .75)^{(2-0)} = 1(1)(.25)^2 = .0625$$

If we get 0 heads, the probability which results in the greatest likelihood of our data is .25

## Coin Flipping

Our goal is to select a  $p$  which results in the highest  $f(x|p)$  given our  $x$

Consider  $x = 1$

$$f(1|.25) = \binom{2}{1} \cdot .25^1 (1 - .25)^{(2-1)} = 2(.25)(.75) = .375$$

$$f(1|.5) = \binom{2}{1} \cdot .5^1 (1 - .5)^{(2-1)} = 2(.5)(.5) = .5$$

$$f(1|.75) = \binom{2}{1} \cdot .75^1 (1 - .75)^{(2-1)} = 2(.75)(.25) = .375$$

If we get 1 heads, the probability which results in the greatest likelihood of our data is .5

Typically we do not limit our parameter space like this (only 3 options), but rather search the entire parameter space for the value which maximizes the likelihood

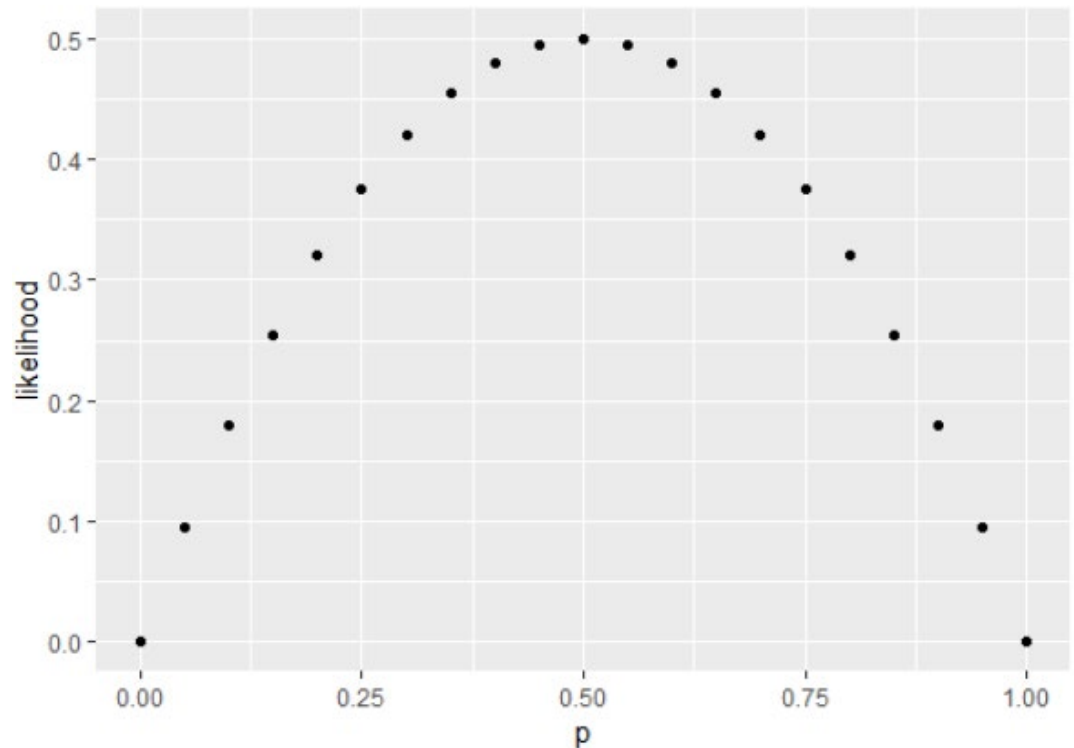
## Finding the Maximum

$$f(x|p) = \binom{2}{x} p^x (1-p)^{(2-x)}$$

Treating  $x$  as fixed (at 1), and varying  $p$  we can represent this likelihood function visually

Visually we can see the maximum is at 0.5

We could also find this by taking the derivative of  $f(x|p)$  with respect to  $p$  and setting that equal to zero



## Thinking about Data

For most of this class we're going to be thinking about data which we assume is normally distributed.

The pdf for the normal distribution looks like this:

$$f(x|\mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

For now, let's assume we know that  $\sigma = 1$

Based on an observed  $x$  we want to get the maximum likelihood estimate of  $\mu$

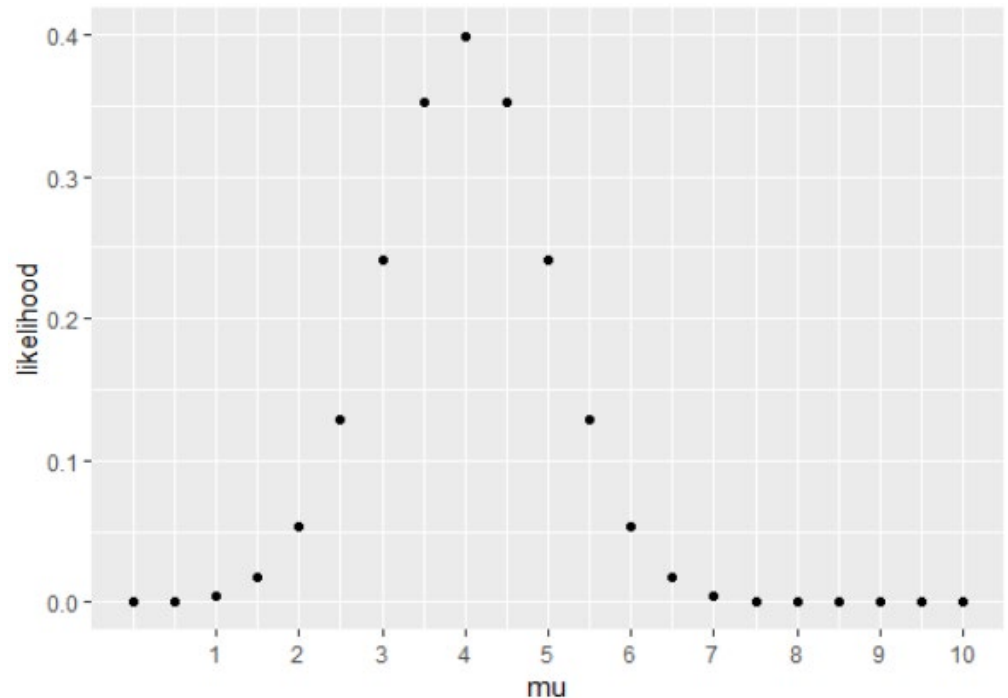
## Finding the Maximum

$$f(x|\mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{(x-\mu)}{\sigma}\right)^2}$$

Let's say that we observe  $x = 4$ , treating  $x$  and  $\sigma$  as fixed, and varying  $\mu$  we can represent this likelihood function visually

Visually we can see the maximum is at 4

We could also find this by taking the derivative of  $f(x|\mu, \sigma)$  with respect to  $\mu$  and setting that equal to zero



## More than 1 value

When we have more than 1 observation on a variable, and those observations are independent, we can calculate the joint likelihood by taking the product of the two likelihoods.

$$f(x_1|\mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{(x_1-\mu)}{\sigma}\right)^2}$$

$$f(x_2|\mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{(x_2-\mu)}{\sigma}\right)^2}$$

$$f(x_3|\mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{(x_3-\mu)}{\sigma}\right)^2}$$

$$f(x_1, x_2, x_3|\mu, \sigma) = f(x_1|\mu, \sigma)f(x_2|\mu, \sigma)f(x_3|\mu, \sigma)$$

This relies on an **independence assumption**

## Deriving the joint likelihood

If we are comfortable assuming that the  $x$ s have the same mean and standard deviation, we can combine their likelihoods

$$\begin{aligned} f(x_1, x_2, x_3 | \mu, \sigma) &= \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{(x_1-\mu)}{\sigma}\right)^2} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{(x_2-\mu)}{\sigma}\right)^2} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{(x_3-\mu)}{\sigma}\right)^2} \\ &= \left(\frac{1}{\sigma\sqrt{2\pi}}\right)^3 e^{-\frac{1}{2}\left(\frac{(x_1-\mu)}{\sigma}\right)^2} e^{-\frac{1}{2}\left(\frac{(x_2-\mu)}{\sigma}\right)^2} e^{-\frac{1}{2}\left(\frac{(x_3-\mu)}{\sigma}\right)^2} \end{aligned}$$

Some old math rules you've probably forgotten:  $e^x e^y = e^{x+y}$

$$\begin{aligned} &= \left(\frac{1}{\sigma\sqrt{2\pi}}\right)^3 e^{-\frac{1}{2}\left(\frac{(x_1-\mu)}{\sigma}\right)^2 + -\frac{1}{2}\left(\frac{(x_2-\mu)}{\sigma}\right)^2 + -\frac{1}{2}\left(\frac{(x_3-\mu)}{\sigma}\right)^2} \\ &= \left(\frac{1}{\sigma\sqrt{2\pi}}\right)^3 e^{-\frac{1}{2\sigma^2}((x_1-\mu)^2 + (x_2-\mu)^2 + (x_3-\mu)^2)} \end{aligned}$$

We want to pick a  $\mu$  which minimizes the squared deviations

## Finding the Maximum

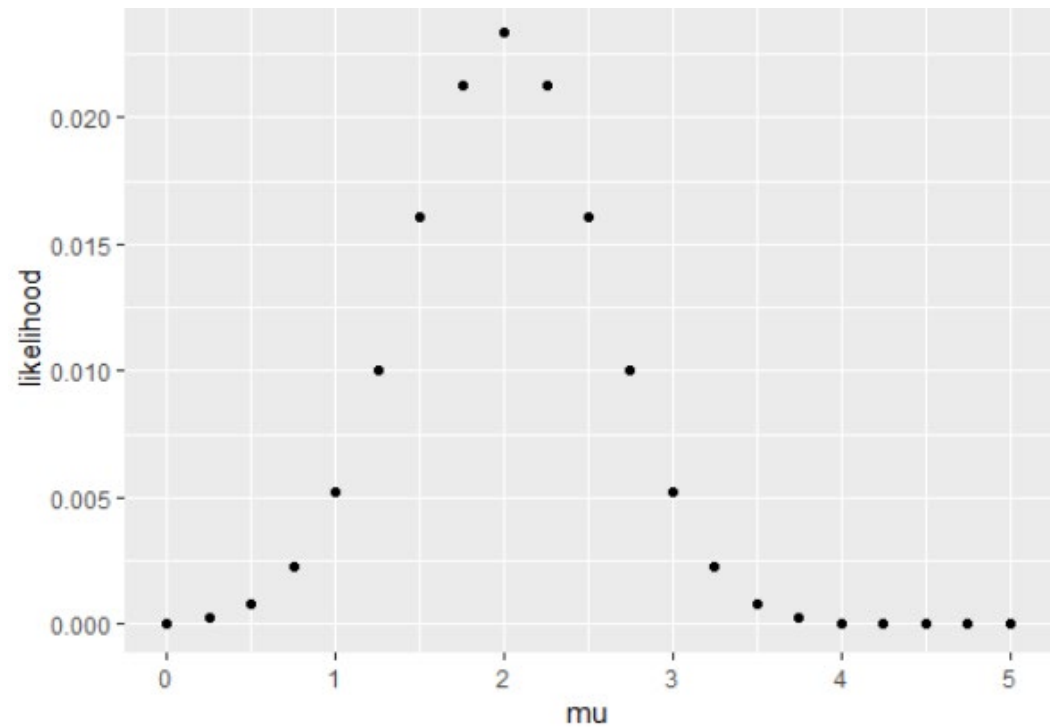
$$f(x_1, x_2, x_3 | \mu, \sigma) = \left( \frac{1}{\sigma\sqrt{2\pi}} \right)^3 e^{-\frac{1}{2\sigma^2}((x_1-\mu)^2 + (x_2-\mu)^2 + (x_3-\mu)^2)}$$

Let's say that we observe  $x = 1, 2, 3$ , treating  $x$  and  $\sigma$  as fixed, and varying  $\mu$  we can represent this likelihood function visually

Visually we can see the maximum is at 2

We could also find this by taking the derivative of  $f(x_1, x_2, x_3 | \mu, \sigma)$  with respect to  $\mu$  and setting that equal to zero

The **sample mean** is the maximum likelihood estimate!



## What about $\sigma$ ?

$$f(x_1, x_2, x_3 | \mu, \sigma) = \left( \frac{1}{\sigma\sqrt{2\pi}} \right)^3 e^{-\frac{1}{2\sigma^2}((x_1-\mu)^2 + (x_2-\mu)^2 + (x_3-\mu)^2)}$$

Let's now treat  $\mathbf{x}$  and  $\mu$  as fixed and vary  $\sigma$

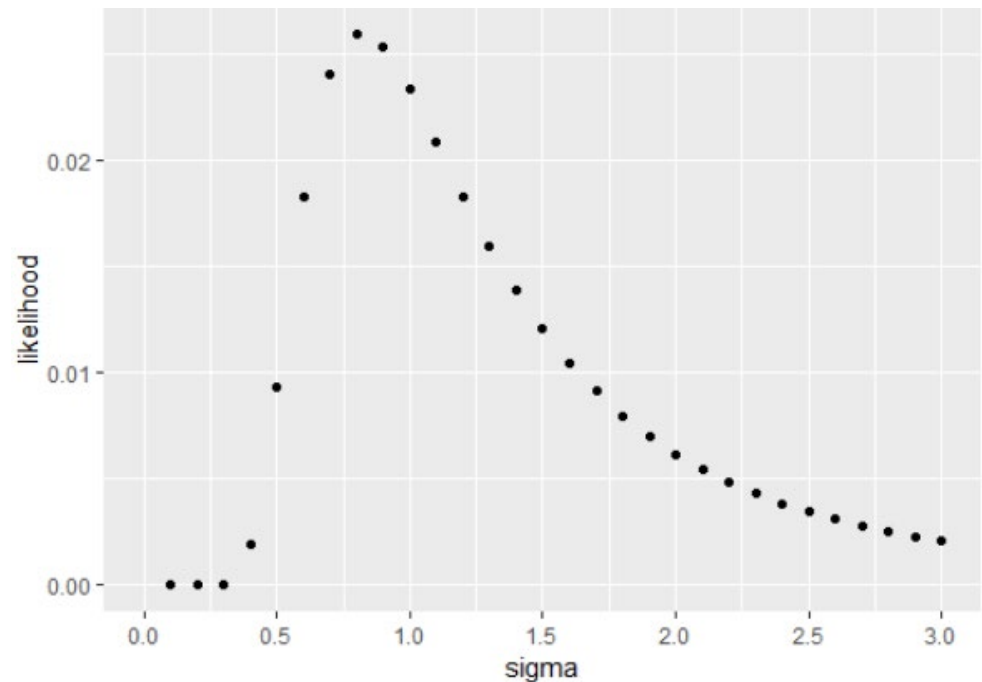
Before we do that what is the sample SD?  $\sqrt{\frac{(1-2)^2 + (2-2)^2 + (3-2)^2}{3-1}} = \sqrt{\frac{1+0+1}{2}} = \sqrt{\frac{2}{2}} = 1$

The maximum is not at 1!

The maximum is at .816. Why?

The population SD estimate (divide by N instead of N-1) is the maximum likelihood estimate of the  $\sigma$

$$\sqrt{\frac{2}{3}} = .816$$



## Multivariate Normal Distribution

In SEM we're focused on the covariance among multiple variables, which we describe as having a **multivariate normal distribution**. Each variable has its own mean ( $\mu_k$ ) and variance ( $\sigma_k^2$ ), but also a covariance with the other variables  $\sigma_{kj}$

All the means together can be expressed as  $\boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \vdots \\ \mu_{N_Z} \end{bmatrix}$

The variances and covariances are saved in a matrix we're already familiar with:

$$\boldsymbol{\Sigma} = \begin{bmatrix} \sigma_1^2 & \cdots & \sigma_{1,N_Z} \\ \vdots & \ddots & \vdots \\ \sigma_{1,N_Z} & \cdots & \sigma_{N_Z}^2 \end{bmatrix}$$

$$f(\mathbf{z}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = (2\pi)^{-(N_Z)/2} \det(\boldsymbol{\Sigma})^{-\frac{1}{2}} e^{-\frac{1}{2}(\mathbf{z}-\boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\mathbf{z}-\boldsymbol{\mu})}$$

# Loglikelihood

Likelihoods can be very small, which makes them hard to work with in computers

Loglikelihoods can be easier to deal with and are monotonic

Logs also have some nice properties

$$\begin{aligned}\log(ab) &= \log(a) + \log(b) \\ \log(a^k) &= k \log(a)\end{aligned}$$

$$f(x_1, x_2, x_3 | \mu, \sigma) = f(x_1 | \mu, \sigma) f(x_2 | \mu, \sigma) f(x_3 | \mu, \sigma)$$

$$\log(f(x_1, x_2, x_3 | \mu, \sigma)) = \log(f(x_1 | \mu, \sigma)) + \log(f(x_2 | \mu, \sigma)) + \log(f(x_3 | \mu, \sigma))$$

Notation is often  $LL(x | \mu, \sigma)$

Sometimes you'll see  $-2LL$  (we want to **minimize** this, rather than maximize it)

## Fitting Function

Another way to think about maximum likelihood is that you define what's called a "fitting function" (like a loss function),  $F(\mathbf{S}, \mathbf{\Sigma}(\boldsymbol{\theta}))$

Fitting function should have these properties:

- It should take a scalar value
- $F(\mathbf{S}, \mathbf{\Sigma}(\boldsymbol{\theta})) > 0$  for all  $\mathbf{S}$  and  $\mathbf{\Sigma}(\boldsymbol{\theta})$
- $F(\mathbf{S}, \mathbf{\Sigma}(\boldsymbol{\theta})) = 0$  if  $\mathbf{\Sigma}(\boldsymbol{\theta}) = \mathbf{S}$
- $F(\mathbf{S}, \mathbf{\Sigma}(\boldsymbol{\theta}))$  is continuous in  $\mathbf{S}$  and  $\mathbf{\Sigma}(\boldsymbol{\theta})$

Any function with these properties would provide a **consistent** estimate of  $\boldsymbol{\theta}$

\*Consistent means converges in probability to the population value

## Maximum Likelihood Fitting Function

$$F_{ML} = \log|\mathbf{\Sigma}(\boldsymbol{\theta})| + \text{tr}(\mathbf{S}\mathbf{\Sigma}(\boldsymbol{\theta})^{-1}) - \log|\mathbf{S}| - (p + q)$$

This function assumes that both  $\Sigma(\theta)$  and  $S$  are positive definite

Assumes  $\mathbf{Z}$  are multivariate normally distributed and  $\mathbf{S}$  has a Wishart distribution

Appendix 4B Bollen (1984) shows how to get from likelihood to fitting function

## Properties of Maximum Likelihood

- Estimates are asymptotically unbiased
- Estimates are consistent
- Estimates are asymptotically efficient
  - Among all consistent estimators none has a smaller asymptotic variance
  - Gives a standard error estimate
- Estimators are asymptotically normal
  - Allows us to use estimate, SE, to get Z-statistic, and  $p$ -value

Other useful properties:

- Scale invariant (rescaling variables doesn't change fit)
- Scale free (rescaling variables changes coefficients in a predictable way)

# Asymptotic Covariance Matrix

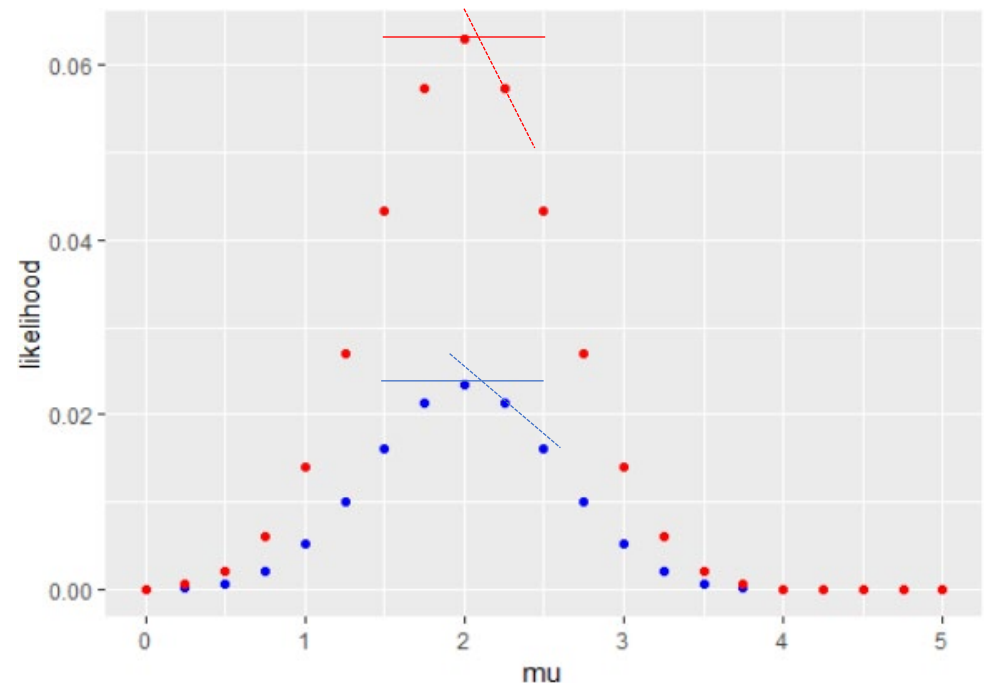
We want estimates of each coefficient, but we also need a numeric representation of the certainty of those estimates.

Consider two possible sets of data:  $x_1 = 1, 2, 3$  vs.  $x_2 = 1.9, 2, 2.1$

Each has the same mean, and thus the same maximum likelihood estimate. But our certainty about that mean should differ.

That certainty can be expressed as degree of change in the slope of the tangent line (i.e., the second derivative of the fit function)

The change in the slope of the tangent is greater for the more closely clustered data (red)



## Asymptotic Covariance Matrix

The asymptotic covariance matrix is calculated from the fitting function

$$\frac{2}{N-1} \left\{ E \left( \frac{d^2 F_{ML}}{d\boldsymbol{\theta} d\boldsymbol{\theta}'} \right) \right\}^{-1}$$

The estimated asymptotic covariance matrix is

$$\frac{2}{N-1} \left\{ E \left( \frac{d^2 F_{ML}}{d\hat{\boldsymbol{\theta}} d\hat{\boldsymbol{\theta}}'} \right) \right\}^{-1}$$

Square root of the diagonal of this matrix are the **standard errors** for the parameters

## Additional Resources

- Quantitude has a nice episode of ML (<https://quantitudepod.org/s4e02-ml/>)
- Hoyle book Chapter 3
- Enders book Chapter 2